

Name Marking Key

Year 9

Investigation 2



Baldivis
Secondary College

Pythagorean Theorem

Time: 50 mins

Marks: /34

1. [10 marks]

a) Square these numbers

b) Square root these numbers

$\frac{1}{2}$ mark each

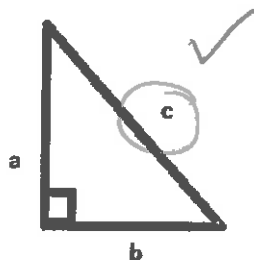
1	1
21	441
3	9
4	16
12	144
29	841
15	225
11	121
20	400
10	100

1	1
2	1.4142...
64	8
36	6
169	13
25	5
16	4
9	3
144	12
225	15

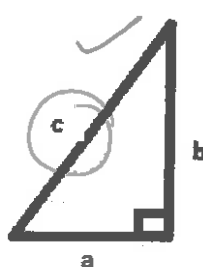
$\frac{1}{2}$ mark each

2. The hypotenuse of a right triangle is always opposite the right angle. Circle the hypotenuse on the following diagrams. [3 marks]

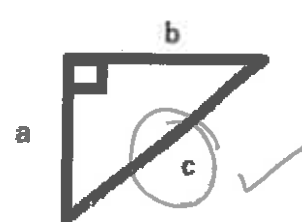
a)



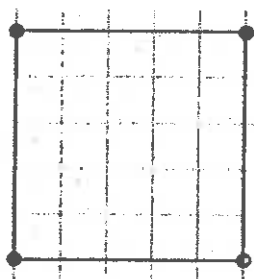
b)



c)



3. Find the area of the following squares. [2 marks]



$$A = 25 \text{ units}^2 \checkmark$$

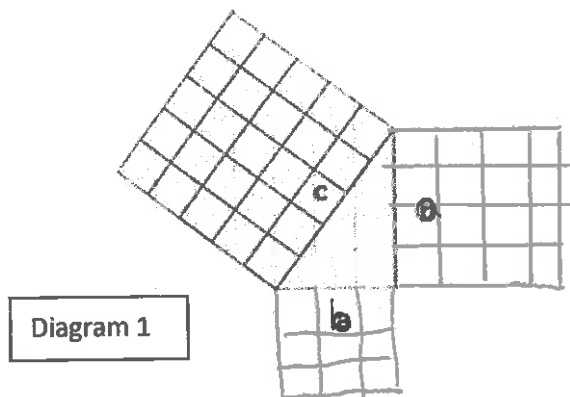


3cm

3cm

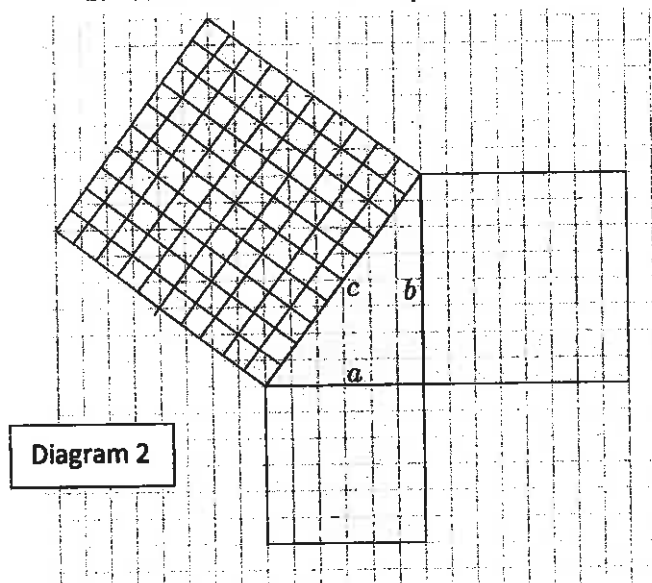
$$A = 9 \text{ cm}^2 \checkmark$$

4. Draw the 4 by 4 square on side a and the 3 by 3 square on side b
- Record the lengths of each of the sides in the table opposite
 - Find the area of each square and record in the table.



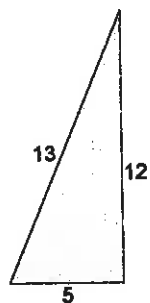
no marks.

- 5.
- Record the lengths of each of the sides in the table opposite
 - Find the area of each square and record in the table.



- 6.
- Record the values in the table

Diagram 3



$\frac{1}{2}$ mark each

	a	b	c	a^2	b^2	c^2
Diagram 1	4	3	5	16	9	25
Diagram 2	6	8	10	36	64	100
Diagram 3	5	12	13	25	144	169

[8 marks]

7. Describe the relationship between a^2 , b^2 and c^2 . [1 marks]

The square of a plus the square of b equals the square of c .
or similar. ✓

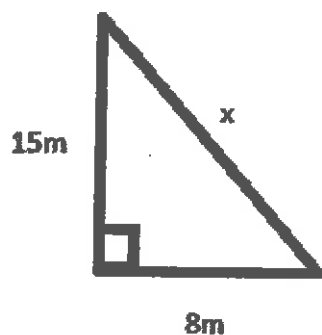
8. Using this relationship fill in the table. [6 marks]

a	b	c	a^2	b^2	c^2
9	12	15	81	144	225
30	16	34	900	256	1156
21	20	29	441	400	841

$\frac{1}{2}$ mark each.

9. Calculate the unknown sides of the right angle triangles. [4 marks]

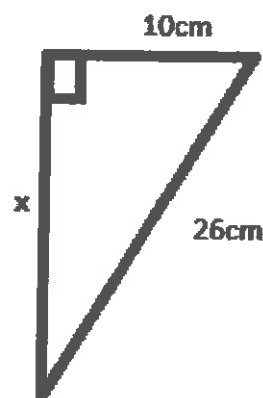
a)



$$x^2 = 289 \checkmark$$

$$x = 17 \text{ cm} \checkmark$$

b)



$$x^2 = 776 \checkmark$$

$$x = 27.86 \text{ cm} \checkmark$$

